

BOOKLET NUMBER:

14947



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VARDHAMAN COLLEGE OF ENGINEERING
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 A8520 22.10.2025 AN

VARDHAMAN
 COLLEGE OF ENGINEERING
 Shamshabad – 501 218, Hyderabad
CAT EXAMINATION ANSWER BOOKLET



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CAT - I / II EXAMINATION I / II / III / IV B.Tech / M.Tech / MBA I / II Semester 10/25
 (Strike off whichever is not applicable) (Month and Year)

Name : Sathvik Rudam
 Branch : CSE
 Subject : Software Engineering
 Subject Code : A8520
 Date of Exam : 22/10/2025

19/22/10/25
 Signature of the Invigilator with date

(Booklet contains 20 pages)

INSTRUCTIONS TO THE CANDIDATES

- This booklet contains 20 pages. Candidates must ensure it before writing and in case a defective answer book is issued it must be returned to the invigilator and a new and defect free booklet must be obtained.
- Before the candidate begins to answer, registered number, particulars of year, semester, subject, etc., are to be filled in. Failure to enter all or any of these particulars may disqualify the paper from valuation.
- Candidate is prohibited from
 - writing:
 - anything addressing the examiner in any manner whatsoever, in their answer book.
 - objectionable/obscene language in the answer book.
 - anything other than their Registered No. on the question paper.
 - either seeking or providing any assistance to the fellow candidates in the exam.
 - possessing a manuscript or a printed matter, in any form, in the examination hall.
 - bringing loose sheets or paper into the examination hall and detaching any paper from the answer book.
 - carrying Mobile Phone, Smart Watches and any other Electronic Gadgets etc., to Exam Hall.

Violation of these instructions will be viewed as a case of malpractice, which is a punishable offence.

- Before beginning to answer any question, candidates must write the correct question number, and should not write
- Answers must be written legibly on both sides of the paper. There shall be about 25 lines in each page. It is not necessary to begin each answer on a fresh page. Candidates should not use any other ink, except BLACK or BLUE ink.
- Rough work, if any, must be separated, from the subject matter, by a line and noted as rough work.
- The answer book, at the end of the examination, must be handed over to the Assistant Superintendent (Invigilator) by the candidate.

This responsibility lies with the candidate only.

- Candidates should maintain absolute silence during the time of examination. Misbehavior, in any form, by the candidate, in the examination hall, will attract severe punishment.
- Candidates are permitted to leave the examination hall only after the allotted time.
- No additional answer books will be supplied.**

To be filled in by the Examiner only

MARKS AWARDED										
PART – A				PART – B						
Q No.1				Q No.	2	3	4	5	6	7
a		f		a						
b		g		b						
c		h		c						
d		i		d						
e		j		Total						
Total (a to j)										
In Words:										
In Figures:										

Signature of the Examiner

START WRITING FROM HERE

Q.No.

PART-A

i) a) Interaction Diagrams show the ~~to~~ interaction b/w components, or user and system, or components to components. Some interaction diagrams are Use Case Diagrams, Sequence Diagrams, etc...

i) b) Object Oriented Design.

i) c) ~~the~~ Structural Diagrams show the structure inside the Software during its processing.

→ Some structural diagrams are Class diagram, Object diagram, etc...

i) d) → System testing refers to testing of whole Software as a system. Here, whole Software collectively is tested including all the modules.

→ Integration testing refers to testing software in repeated iterative model. Testing it again and again. Testing Software by continuously adding new parts until whole software is ready.

~~the~~)

i) e) → Top-down integration refers to testing Software top from the driver code to the bottom.

Ex: Testing ERP module report module.

It first starts from login, fetches students details, and generates ~~the~~ report.

Q.No.

- ✂ 1) f) → Primary Purpose of debugging is to make sure that no errors (or) bugs are left in the software so it runs smoothly.
→ Primary purpose is to make the software error-free & efficient.
- 1) g) First step in Strategic approach Testing is:
Branch testing: i.e. testing if any conditional ~~st~~it exists, then each way should be run atleast once.
- 1) h) → Formal Technical Reviews is an activity of Software Quality Analysis where the documents of software are checked thoroughly to find any error before deployment.
- ✂ 1) i) Yes, failure costs are higher than appraisal costs because it is difficult to correct a software after deployment ~~than~~ at development stage. It costs more at finish stage.
- 1) j) → It ensures by regulating all companies that their product must meet the users expectations, needs, and quality. The ISO also focuses on improving quality thoroughly. In this way, it influences customer satisfaction.
- ✂

Q.No.

Part-B

2) ~~It~~, It is very important to make the user interface stay consistent. It is also one of the Golden rules.
"Make the interface consistent".

→ Making the user interface consistent helps the user simplify in identifying everything. If the interface is inconsistent, then the user will struggle very much to find required buttons, options, and many more.

→ If in a web interface, all the pages are consistent, i.e. uses same interface style, it is easier for user to navigate and improves usability and experience.

→ Instead, if the submit button for a form is placed somewhere below the fields, user wouldn't be able to find it.

Ex: ERP login system for College.

→ Consider college ERP login system, where if the login button is placed at some other place than at bottom corner below user & pass, then the user would not be able to find it.

→ If in one page the button is at the bottom of page, and in the next page if it is at top, the user might feel disappointed & struggle to find it.

→ So, all the pages should be consistent with one design.

Q.No.

→ This helps in improving user experience.

Ex: In a shopping website, you will be adding products to ~~cards~~ and visiting the cart. But, if the place order button is placed at some other place, then it would reduce user experience.

→ Instead, making a consistent interface would be more efficient even if it is not perfect.

→ Some examples of consistent interfaces are:

- 1) Google: every page has the same styled interface.
- 2) Shopping sites: All shopping websites follow a standard style which is very convenient to users.

Q.No.

3) → Use case Diagram:

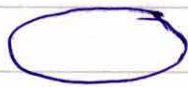
- Use Case Diagram is a type of Unified Modelling Language Diagram.
- It represents the interaction between the User and the system.
- Use Case diagram is used to represent Interaction models.
- Use Case diagram comes under behavioral diagrams.
- Use case diagram is represented by using:



→ user (or) character



→ link (or) association.



→ process (or) thing happening.

→ the data or task is written inside the oval to represent the task.

→ the user symbol is called as stick man.

→ link symbol is called as stick.

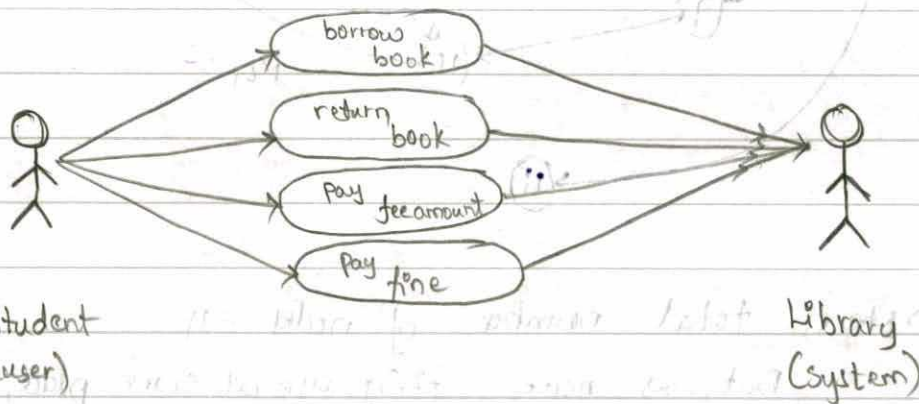
→ Use Case Diagram for Library Management System:

- Let us consider some cases where the user interacts with the system (Library):

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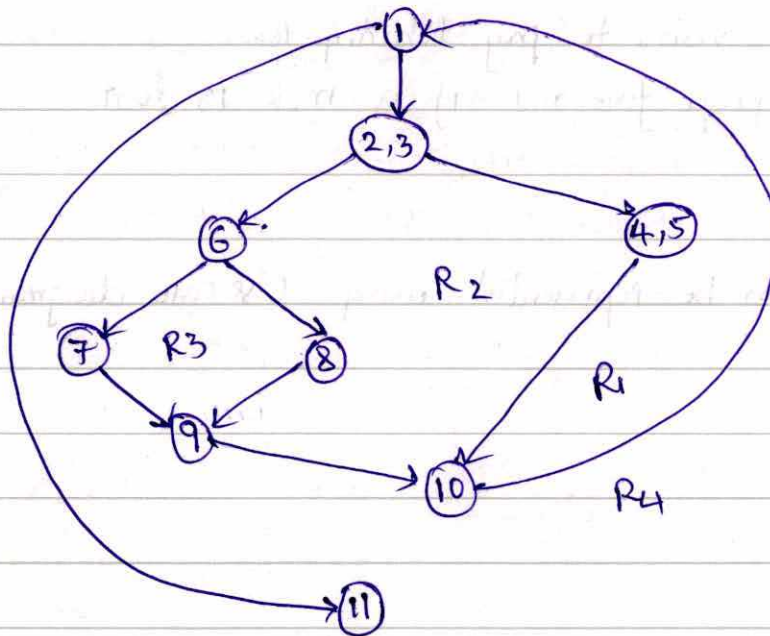
- i) User comes to borrow book
- ii) User visits to return book
- iii) User visits to pay library fee
- iv) User pays fine amt if a book is lost

This can be represented using Use case diagram as:



Q.No.

4) Given Graph:



→ Here, total number of nodes = 11

But, as nodes {2,3} are at same place, both are considered as one

nodes {4,5} also considered ^{as} one.

∴ Total no. of nodes are = 9

Now, total no. of edges = 11

They are: {1 → 2,3}, {2,3 → 4,5}, {2,3 → 6},
{6 → 7}, {6 → 8}, {8 → 9}, {7 → 9},
{4,5 → 10}, {9 → 10}, {10 → 11}, {1 → 11}

→ Now, formula to find total no. of independent paths are:

Q.No.

$$V(G) = E - n + 2$$

where; E = edges of graphs

n = nodes of graphs

$$\Rightarrow \text{total independent paths} = 11 - 9 + 2 \Rightarrow 4$$

$\therefore$ Total independent paths, 4 exists.

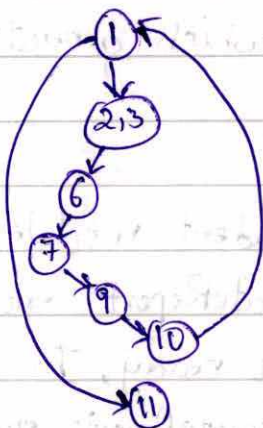
$\rightarrow$ The paths are:

$\{1 \rightarrow 2, 3 \rightarrow 6 \rightarrow 7 \rightarrow 9 \rightarrow 10 \rightarrow 11\}$,

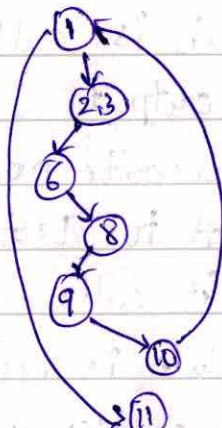
$\{1 \rightarrow 2, 3 \rightarrow 6 \rightarrow 8 \rightarrow 9 \rightarrow 10 \rightarrow 11\}$,

$\{1 \rightarrow 2, 3 \rightarrow 4, 5 \rightarrow 10 \rightarrow 11\}$, and

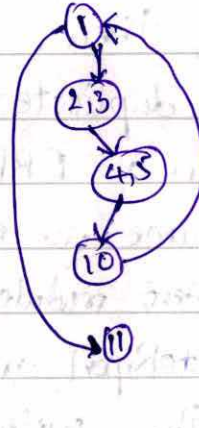
$\{1 \rightarrow 11\}$.



Path-1



Path-2



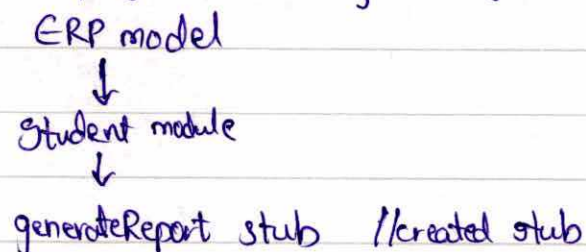
Path-3



Path-4

Q.No.

- 5) → stub and driver programmings are part of testing.
 → Stub and driver are mostly used while performing unit testing.
 → When we want to test a code in a software, but this module redirects to another part of code which is not yet ready.
 → At this case, stub is used.
 → Stub is like a duplicate sample code which gives sample output when a code we are testing calls it.
 → When the module ~~are~~ we are running in SW is being tested, if this module should call another module, then stub is called which generates fake/duplicate output.
 Ex: In a ERP website model,
 • assume we need to test student module, but in student module it calls generateReport module.
 generateReport module is not yet ready, In this case, the system creates a generateReport stub which generates fake output when student module calls it.
 In this way we can perform testing errorfree.



Q.No.

Driver programming is also mostly similar to stub, but instead it acts as driver code.

→ Assume we are testing a module which needs to be called by a module. But, the parent module is not yet ready.

→ In this case, we use the driver module.

→ Driver is like a duplicate starter code, which calls the module the needs to be tested.

→ When we try to test required module, driver automatically calls the module that we are testing.

Ex:

• Let us assume if we need to test a product page in a shopping website. But it can't be called without home interface. In this situation:

• A driver is created and it calls the product page which we are testing.

Driver home page
(shopping website)



Product page

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6)

7)

→ The main elements (or) promises of ISO 9001 quality management system are:

- i) Ensure Quality Adherence
- ii) Ensures User Satisfaction
- iii) Continuous Improvement
- iv) Regulates Companies.

i) Ensure Quality Adherence:

- It ensures that the organizations must maintain high (or) expected quality by user.
- The product must meet user's expectations.
- All the documentation of the product (software) should be perfect and reviewed thoroughly.

ii) Ensure User Satisfaction:

- The ISO 9001 focuses mainly on user satisfaction.
- It ensures and tries it's best to make the user satisfied with the product
- The product must reach the user's expectations in all ways.

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iii) Continuous Improvement:

- The ISO focuses on continuous improvement.
- It tries continuously to improvise itself in every aspect for every update, to match user requirements.

iv) Regulates Companies:

- The ISO regulates all the companies with strict rules to follow
- It regulates with thorough document check, perfect product (software) and many more.

→ In this way, the ISO 9001's elements ensure product and process quality in SW development.

Q.No.

1. What is the importance of the following factors in the selection of a material?
 (a) Cost
 (b) Availability
 (c) Processing
 (d) Performance
 (e) Environmental

2. What are the factors that affect the selection of a material?
 (a) Cost
 (b) Availability
 (c) Processing
 (d) Performance
 (e) Environmental

3. What are the factors that affect the selection of a material?
 (a) Cost
 (b) Availability
 (c) Processing
 (d) Performance
 (e) Environmental

